Chapter 04 Raw Textbook Content

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4. Basic Set-Up of Full-Endoscopic Spine Surgery

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4.1 Introduction

Full-endoscopic spine surgery (FESS) is an ultra-minimally invasive technique that utilizes an endoscope to access and treat various spinal pathologies. This approach offers reduced tissue trauma, quicker recovery times, and improved visualization of the operative field compared to traditional open techniques, with comparable safety and efficacy [1–3].

While various nomenclature for endoscopic procedures in spine surgery exist, the term “full-endoscopic spine surgery” is defined as that which is performed through the working channel that is within the endoscope itself [4, 5]. Also known as uniportal endoscopy, this distinguishes itself from other techniques such as unilateral biportal

endoscopy (UBE), in which the instruments and endoscope are passed through different portals. In FESS, a working sheath is inserted over serial dilators, allowing for access to the spinal canal through incisions of 1 cm or less. This allows for direct access to the pathology while preserving the surrounding normal tissue as much as possible. The continuous irrigation of fluid, together with high-definition camera systems now available, also allows for excellent visualization of the structures in the spinal canal. In the lumbar spine, there are two main approaches involved in FESS: (i) the transforaminal approach and (ii) the interlaminar approach. The success of FESS relies heavily on meticulous preparation and proper set-up of both the surgical environment and the specialized equipment required for these procedures. This chapter aims to detail the basic set-up of FESS procedures, particularly in lumbar spine surgery.

4.2 FESS Equipment

The basic equipment required for any FESS procedure is the endoscope with its light source, a camera and TV system, normal saline irrigation fluid, and C-arm imaging for localization. It is recommended to use an irrigation pump system specially designed for endoscopic spine procedures in order to ensure continuous and regulated irrigation of fluid. Alternatively, gravity-assisted saline inflow may be used, and a 3000-cc normal saline bag can be placed 50–75 cm above the level of the operating table.